

Varshith K

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AI/ML Engineer | 3+ Years of Experience

Why Am I The Best For This Role

AI/ML Engineer with 3+ years of experience designing, deploying, and optimizing data-driven AI systems across healthcare, supply chain, and enterprise automation. Strong background in LLM engineering, data science, and MLOps — integrating tools like **LangChain, GPT-4 API, Pinecone, TensorFlow, PyTorch, Spark, and Kubernetes** to deliver scalable, production-grade ML solutions. Adept at **troubleshooting real-time data issues**, improving inference latency, and ensuring system reliability through observability, retraining, and bias monitoring. Passionate about bridging research innovation with business outcomes.

Technical Skills

AI/LLM Frameworks: LangChain, LangGraph, MemGPT, GPT-4 API, Whisper, ElevenLabs, Pipecat, LiveKit

Machine Learning: TensorFlow, PyTorch, Scikit-learn, XGBoost, Random Forest, NLP, RAG, Vector Embeddings

Data & ETL: Spark, SQL, Pandas, NumPy, Pinecone, Redis, JSON/CSV Pipelines

MLOps & Infra: Docker, Kubernetes, MLflow, Terraform, Airflow, CI/CD, AWS, Azure, GCP

Backend & APIs: FastAPI, REST, WebSocket, Celery, JWT Authentication

Visualization & Reporting: Tableau, Power BI, Grafana, Prometheus

Programming: Python, R, Bash, Git

Certifications

- [AWS Certified Data Engineer – Associate](#)
 - [Generative AI Fundamentals – Databricks](#)
 - [Tableau 2024.1: Essential Training](#)
 - [Excelerate Data Analyst Associate Certificate](#)
 - AWS Certified AI Practitioner (In Progress)
 - AWS Certified Machine Learning Engineer – Associate (In Progress)
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Professional Experience

Cardinal Health – Dublin, OH

AI Engineer | Oct 2024 – Present

- Architected a **multi-agent AI calendar management platform** using *LangChain, LangGraph, and MemGPT*, orchestrating 8 specialized agents for tasks, calls, and reminders with **memory-based context switching**.
 - Designed **voice-to-intent pipeline** integrating *Whisper, ElevenLabs, and Twilio*, enabling outbound calls, real-time transcription, and WhatsApp automation.
 - Built a **fault-tolerant backend** with *Celery, Redis, and WebSockets* for seamless concurrency, integrating **JWT-based authentication** for user isolation.
 - Developed **RAG-powered document automation** with *GPT-4 embeddings and Pinecone*, improving retrieval accuracy by 40% through tuned chunking and keyword-based scoring.
 - Engineered **self-healing workflows** using retry queues and dead-letter routing, cutting manual incident interventions by 95%.
 - Implemented **observability dashboards** in *Prometheus/Grafana* for latency and memory metrics, proactively detecting bottlenecks before user impact.
 - Led troubleshooting during a live deployment where API rate-limiting caused message delays — introduced **asynchronous batching and circuit breakers**, reducing failure retries by 70%.
 - Collaborated cross-functionally to convert ambiguous product ideas into deployable AI features, delivering **two product launches ahead of schedule**.
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Infosys – Bengaluru, India

AI/ML Engineer | Feb 2022 – Jul 2023

- Led **data-driven model development** for healthcare analytics and logistics optimization using *XGBoost*, *Random Forest*, and *PySpark*.
- Built **risk prediction models** that reduced false positives by 18%, improving patient outcome forecasting accuracy.
- Automated **data drift detection and retraining pipelines**, ensuring long-term model accuracy on dynamic data streams.
- Designed **Spark ETL pipelines** that processed millions of patient and inventory records, cutting ingestion time by ~40%.
- Deployed production ML models on **AWS SageMaker and Azure ML**, implementing **CI/CD templates** to reduce release cycles by 30%.
- Troubleshoot high GPU utilization in inference endpoints — optimized quantization and batch sizing to cut latency by 35%.
- Built **bias monitoring dashboards** and implemented fairness constraints to align with Responsible AI principles.
- Integrated predictive insights into BI dashboards (*Tableau*, *Power BI*), enabling data-backed decisions for 20+ enterprise users.
- Created **custom ML utility packages** for feature engineering and evaluation, reducing team development time by 20%.

Aadya Informatics Private Limited – Hyderabad, India

Data Scientist | Jan 2021 – Jan 2022

- Designed **data validation pipelines** that flagged anomalies and missing data, improving dataset quality by 25%.
- Automated **environment provisioning** using *Terraform*, ensuring reproducible ML infrastructure and reducing setup time by 50%.
- Implemented **blue-green and canary deployments** for ML rollouts, achieving zero downtime during version upgrades.
- Integrated *MLflow* and *TensorBoard* for experiment tracking and model versioning, improving traceability and reproducibility.
- Managed **Kubernetes clusters for ML workloads**, optimizing scheduling and autoscaling to cut cloud costs by 15%.
- Built **Prometheus-based alerts** to monitor drift and model health; proactively mitigated a production degradation by re-triggering retraining on flagged anomalies.
- Deployed **CI/CD pipelines for ML models** ensuring consistent build, test, and deployment across environments.
- Established **data governance and lineage tracking** to ensure compliance with privacy standards and audit readiness.

Education

George Mason University – Fairfax, VA

Master of Science, Data Analytics Engineering | Aug 2023 – May 2025

BITS Pilani – Hyderabad, India

Bachelor of Engineering, Mechanical Engineering | Aug 2019 – May 2023
